BANK DATABASE – ACID PROPERTIES & ISOLATION

2520030071

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Create Database

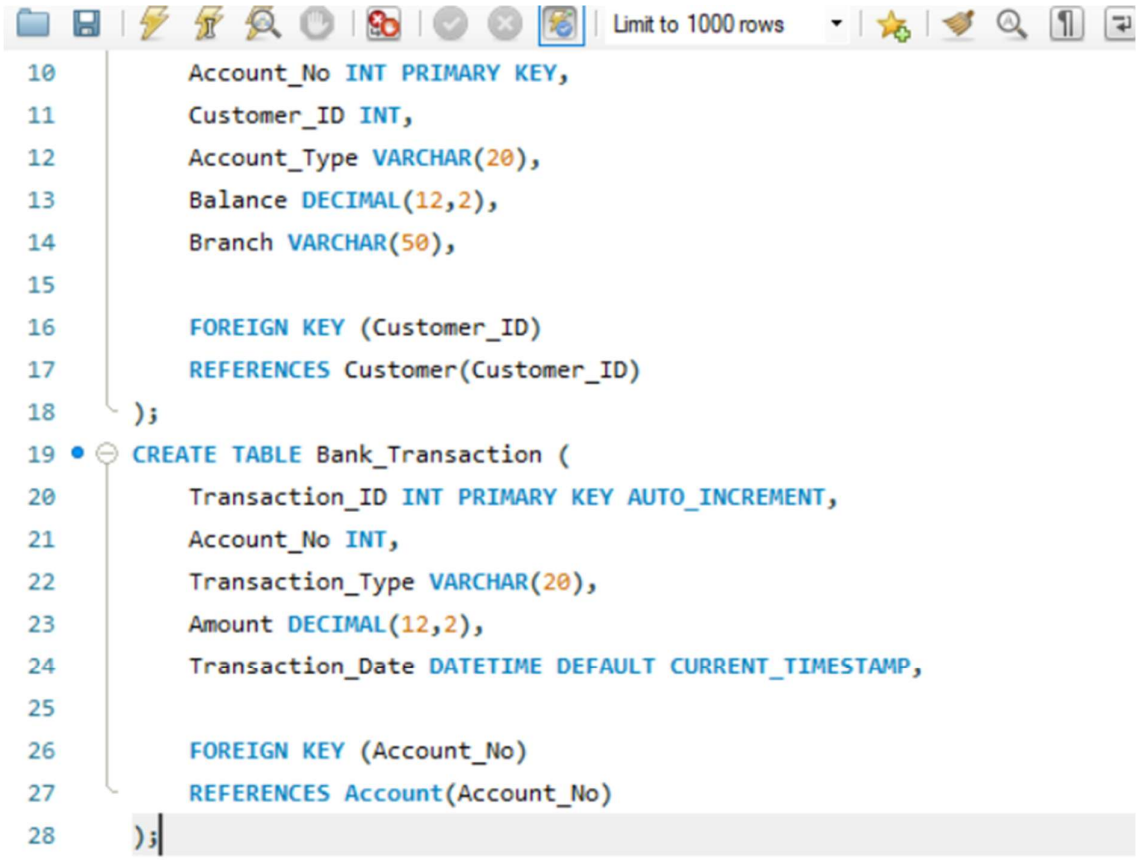
```
1 • CREATE DATABASE Bank_ACID_DB;
2 • USE Bank_ACID_DB;
```

Output

Action Output

#	Time	Action	Message
1	19:24:31	CREATE DATABASE Bank_ACID_DB	1 row(s) affected
2	19:24:37	USE Bank_ACID_DB	0 row(s) affected

Create Customer Table , Accounts Table and Bank Transaction Table



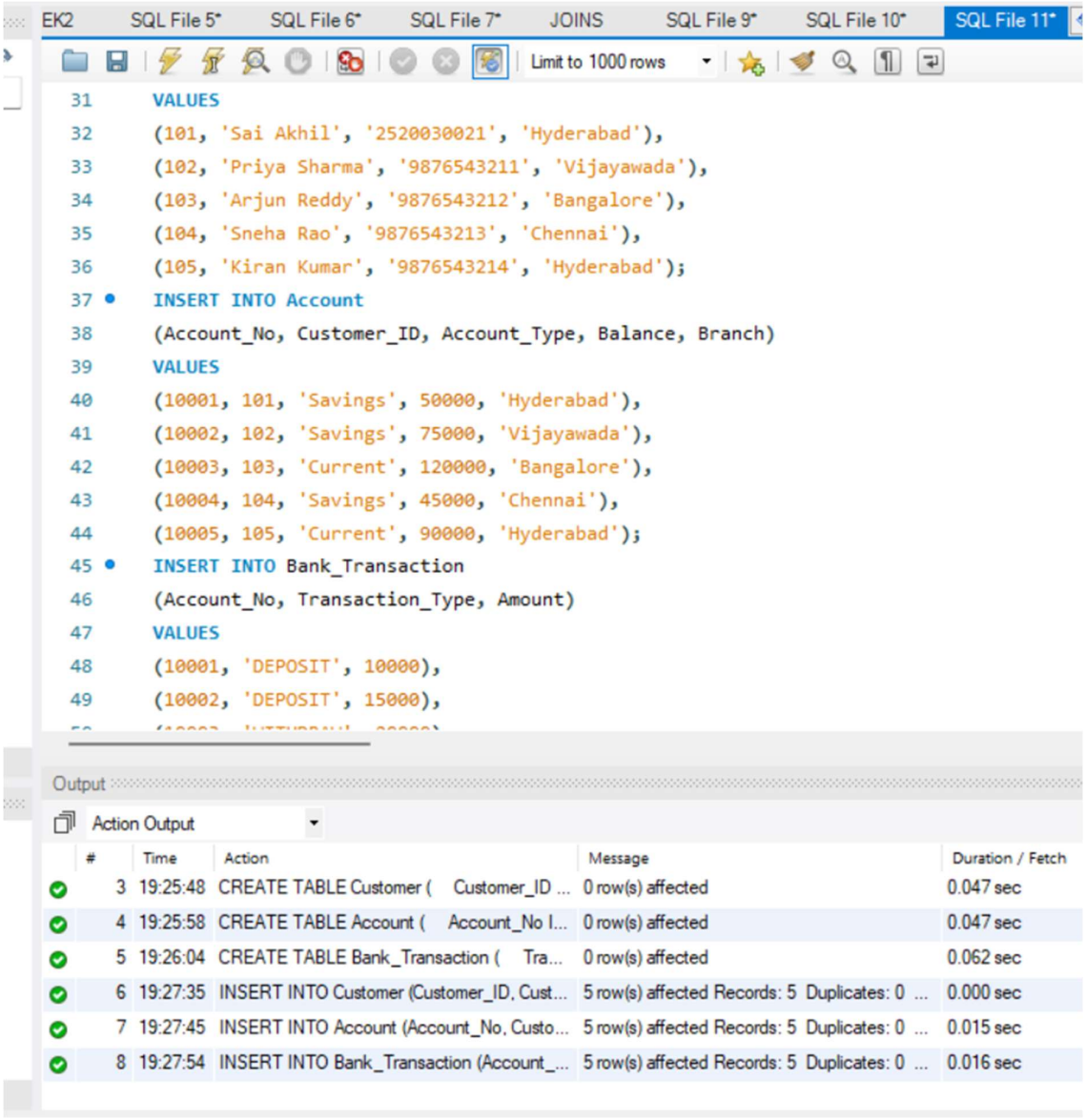
```
10      Account_No INT PRIMARY KEY,
11      Customer_ID INT,
12      Account_Type VARCHAR(20),
13      Balance DECIMAL(12,2),
14      Branch VARCHAR(50),
15
16      FOREIGN KEY (Customer_ID)
17      REFERENCES Customer(Customer_ID)
18  );
19  CREATE TABLE Bank_Transaction (
20      Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
21      Account_No INT,
22      Transaction_Type VARCHAR(20),
23      Amount DECIMAL(12,2),
24      Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
25
26      FOREIGN KEY (Account_No)
27      REFERENCES Account(Account_No)
28  );
```

Output

Action Output

#	Time	Action	Message
1	19:24:31	CREATE DATABASE Bank_ACID_DB	1 row(s) affected
2	19:24:37	USE Bank_ACID_DB	0 row(s) affected
3	19:25:48	CREATE TABLE Customer (Customer_ID I...	0 row(s) affected
4	19:25:58	CREATE TABLE Account (Account_No IN...	0 row(s) affected
5	19:26:04	CREATE TABLE Bank_Transaction (Trans...	0 row(s) affected

Insert Customer Data,Insert Account Data ,Insert Bank Transaction Data



Display Data

```
53 •  SELECT * FROM Customer;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Customer_ID	Customer_Name	Phone	City
101	Sai Akhil	2520030021	Hyderabad
102	Priya Sharma	9876543211	Vijayawada
103	Arjun Reddy	9876543212	Bangalore
104	Sneha Rao	9876543213	Chennai
105	Kiran Kumar	9876543214	Hyderabad
NULL	NULL	NULL	NULL

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utput

Action Output

#	Time	Action	Message
4	19:25:58	CREATE TABLE Account (Account_No...	0 row(s) affected
5	19:26:04	CREATE TABLE Bank_Transaction (Tr...	0 row(s) affected
6	19:27:35	INSERT INTO Customer (Customer_ID, Cu...	5 row(s) affected Records: 5 Duplicates: 0
7	19:27:45	INSERT INTO Account (Account_No, Cust...	5 row(s) affected Records: 5 Duplicates: 0
8	19:27:54	INSERT INTO Bank_Transaction (Account...	5 row(s) affected Records: 5 Duplicates: 0
9	19:28:38	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned

```
55 •  SELECT * FROM Account;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

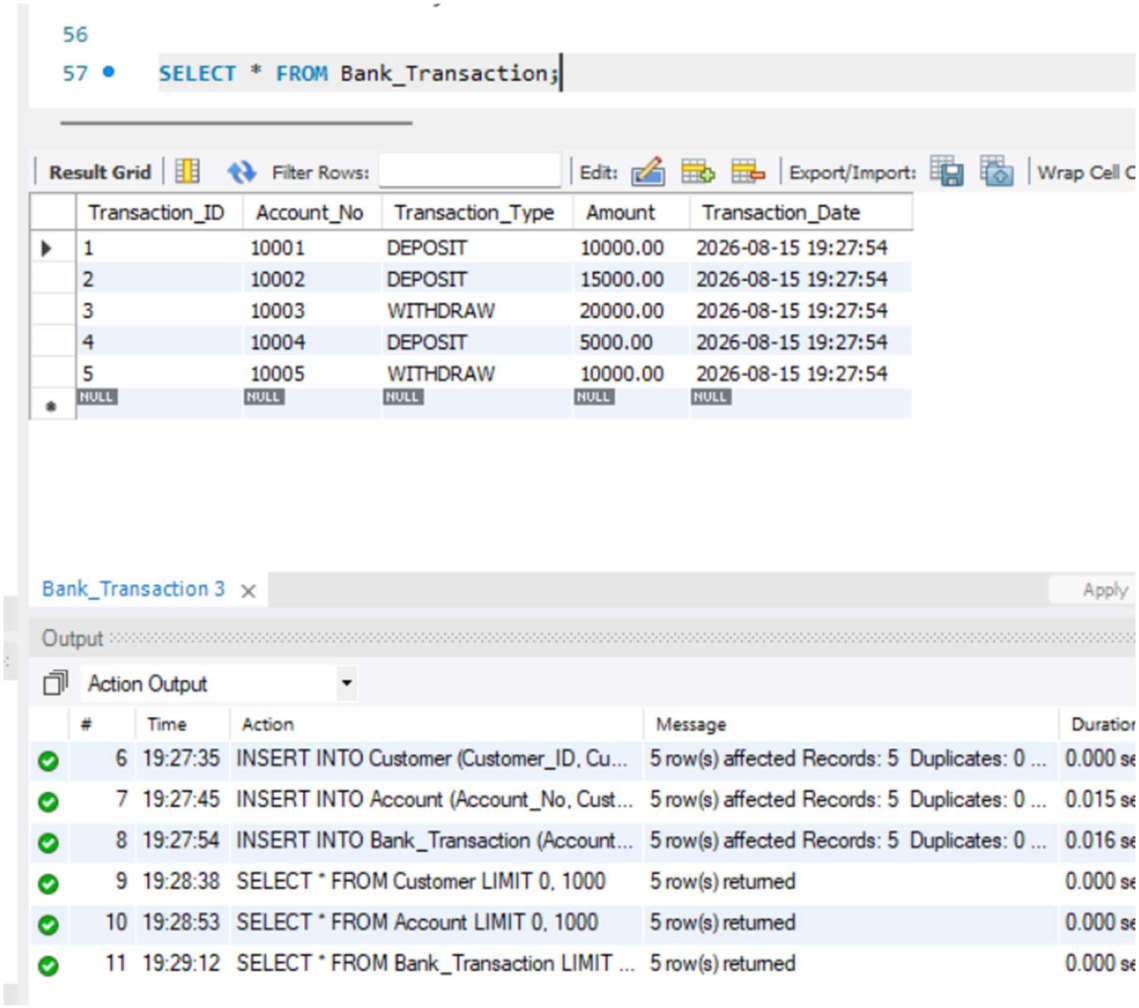
Current

Account 2

Output

Action Output

#	Time	Action	Message
5	19:26:04	CREATE TABLE Bank_Transaction (Tr...	0 row(s) affected
6	19:27:35	INSERT INTO Customer (Customer_ID, Cu...	5 row(s) affected Records: 5 Duplicates: 0 ...
7	19:27:45	INSERT INTO Account (Account_No, Cust...	5 row(s) affected Records: 5 Duplicates: 0 ...
8	19:27:54	INSERT INTO Bank_Transaction (Account...	5 row(s) affected Records: 5 Duplicates: 0 ...
9	19:28:38	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	19:28:53	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned



Start Transaction

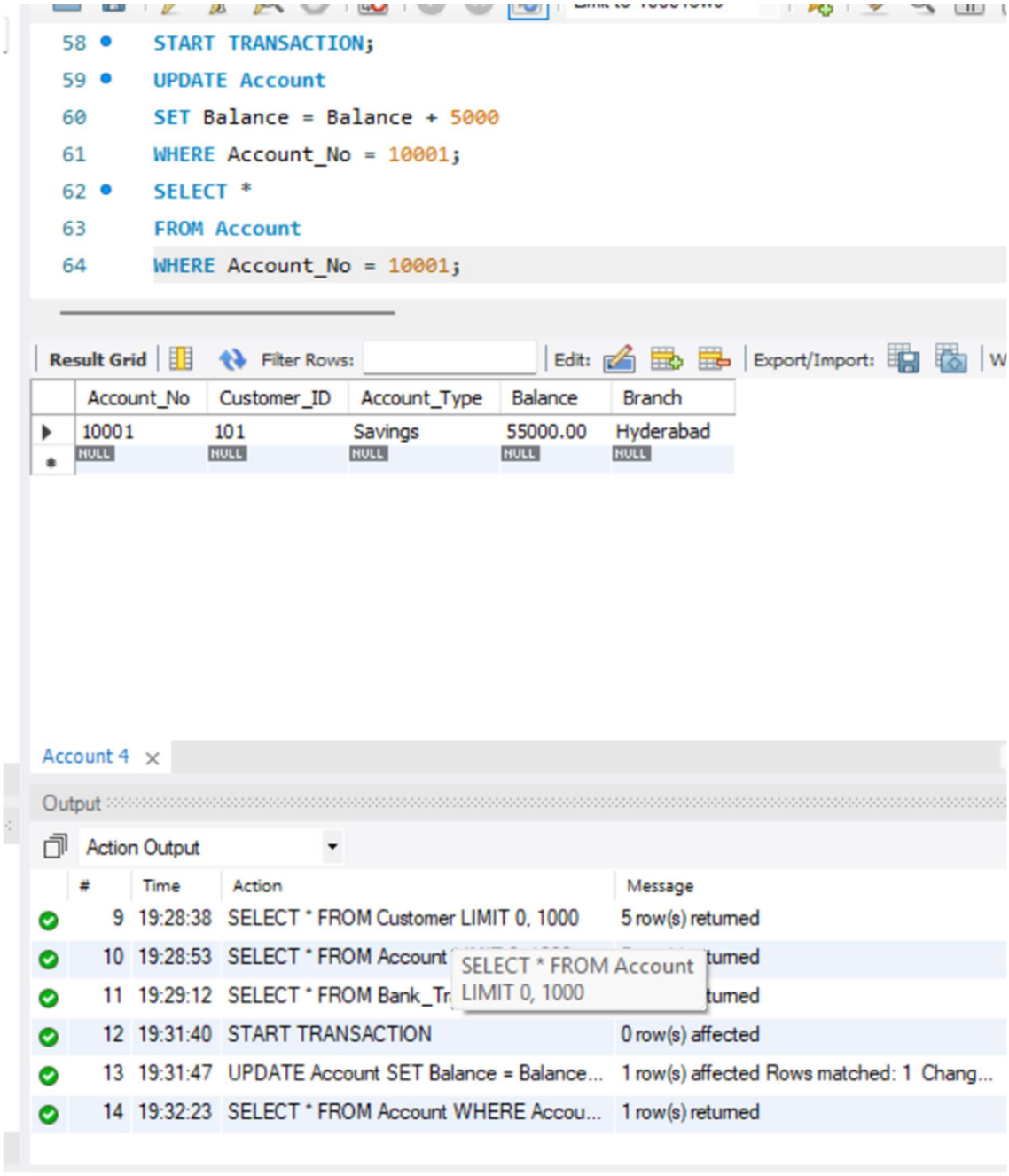
```
58 •  START TRANSACTION;
59 •  UPDATE Account
60    SET Balance = Balance + 5000
61    WHERE Account_No = 10001;
```

Output

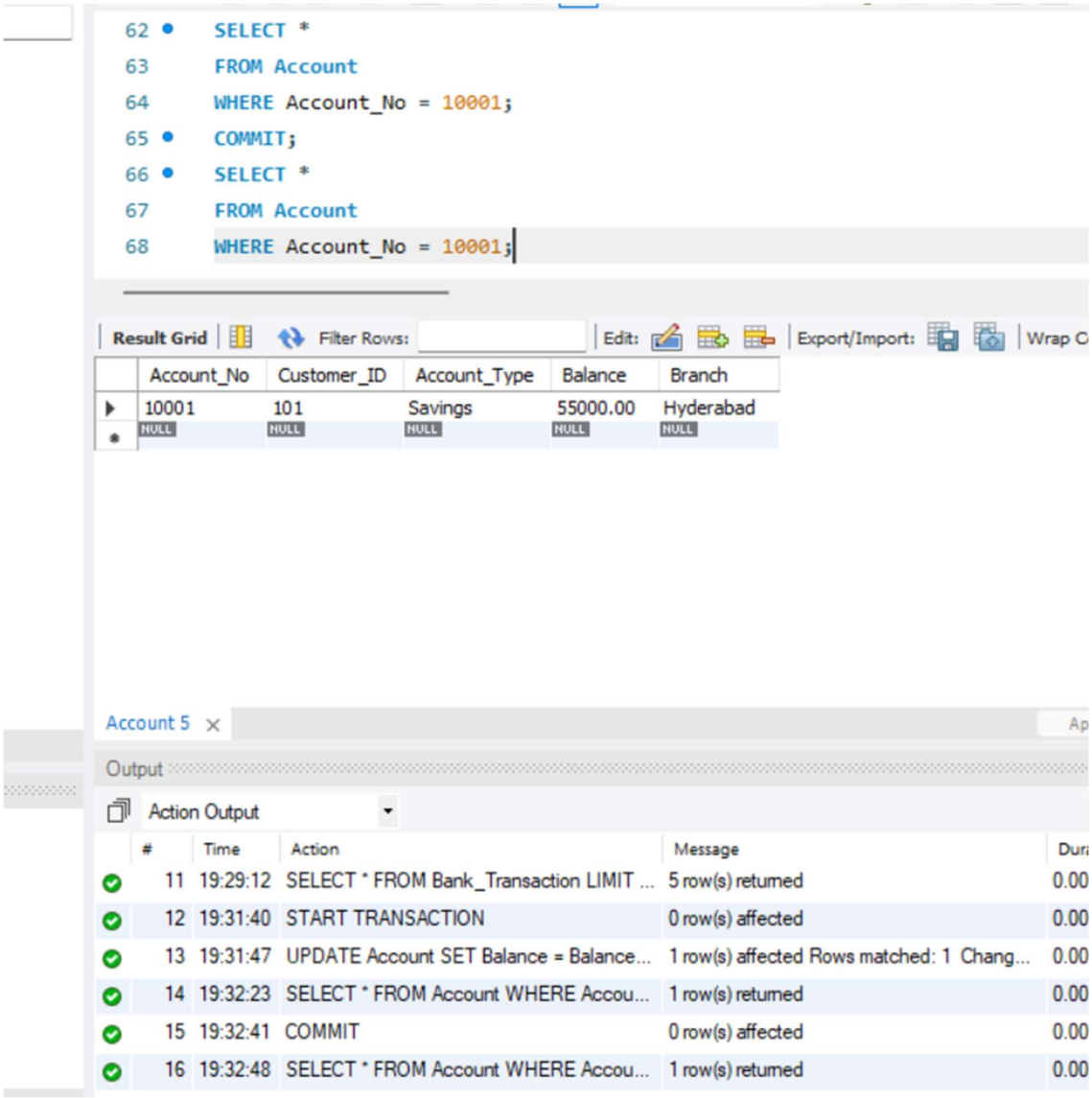
Action Output

#	Time	Action	Message
8	19:27:54	INSERT INTO Bank_Transaction (Account...	5 row(s) affected Records: 5 Duplicate
9	19:28:38	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	19:28:53	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	19:29:12	SELECT * FROM Bank_Transaction LIMIT ...	5 row(s) returned
12	19:31:40	START TRANSACTION	0 row(s) affected
13	19:31:47	UPDATE Account SET Balance = Balance...	1 row(s) affected Rows matched: 1 Ch

Check Balance



Commit and Check Transaction After commit



Start Transaction and Withdraw 10,000

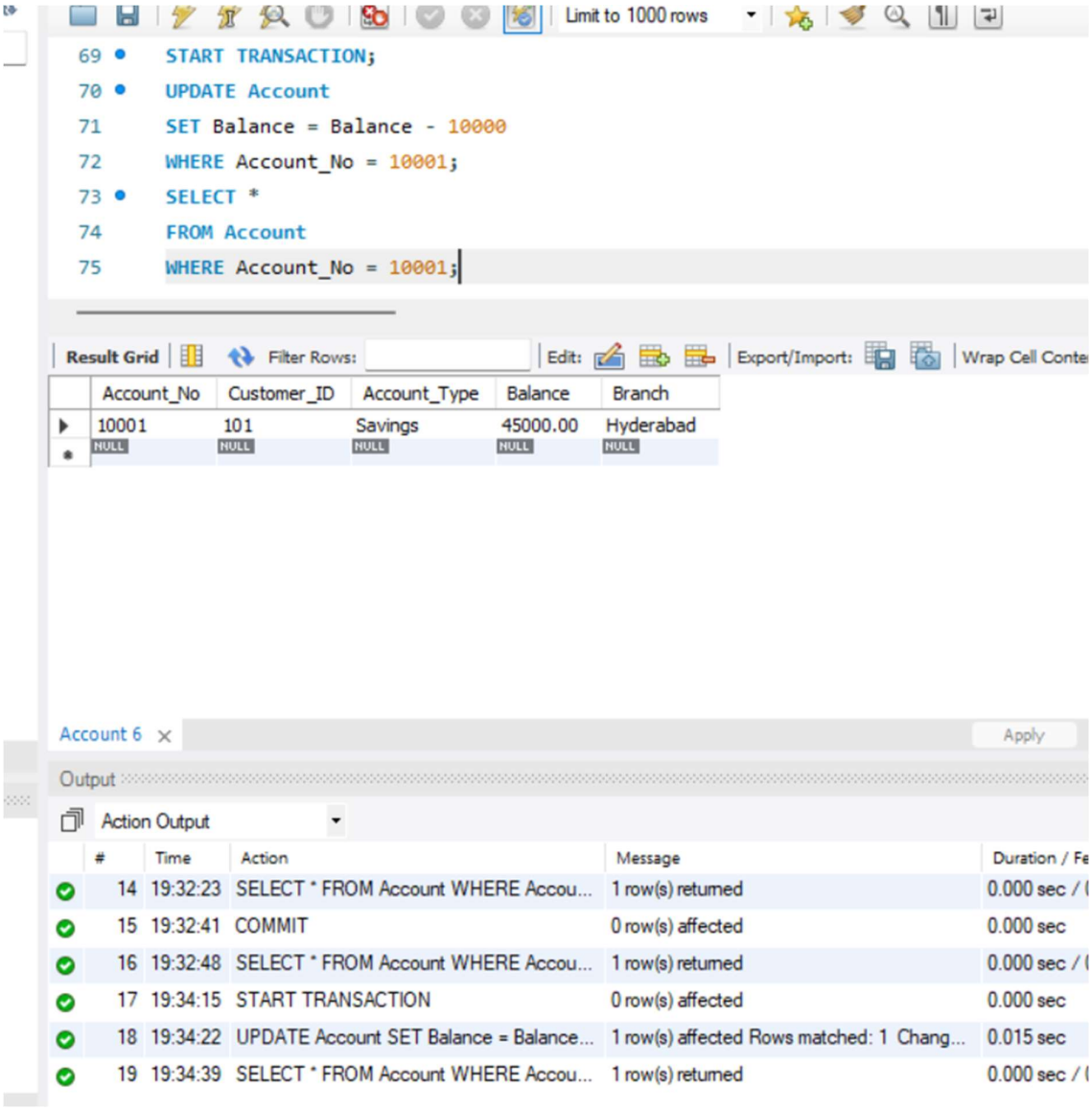
```
69 •  START TRANSACTION;
70 •  UPDATE Account
71    SET Balance = Balance - 10000
72    WHERE Account_No = 10001;
```

Output

Action Output

#	Time	Action	Message
13	19:31:47	UPDATE Account SET Balance = Balance...	1 row(s) affected Rows matched: 1 Chang...
14	19:32:23	SELECT * FROM Account WHERE Accou...	1 row(s) returned
15	19:32:41	COMMIT	0 row(s) affected
16	19:32:48	SELECT * FROM Account WHERE Accou...	1 row(s) returned
17	19:34:15	START TRANSACTION	0 row(s) affected
18	19:34:22	UPDATE Account SET Balance = Balance...	1 row(s) affected Rows matched: 1 Chang...

Check Balance



RollBack and Check Balance after rollback

```
76 •  ROLLBACK;
77 •  SELECT *
78    FROM Account
79    WHERE Account_No = 10001;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Conte

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Account 7 × Apply

Output

Action Output

#	Time	Action	Message	Duration / Fe
15	19:32:41	COMMIT	0 row(s) affected	0.000 sec
16	19:32:48	SELECT * FROM Account WHERE Accou...	1 row(s) returned	0.000 sec / (
17	19:34:15	START TRANSACTION	0 row(s) affected	0.000 sec
18	19:34:22	UPDATE Account SET Balance = Balance...	1 row(s) affected Rows matched: 1 Chang...	0.015 sec
19	19:34:39	SELECT * FROM A	etumed	0.000 sec / (
20	19:35:15	SELECT * FROM A	etumed	0.000 sec / (

```
UPDATE Account
SET Balance = Balance - 10000
WHERE Account_No = 10001
```